

STOCKHOLMN BROMMA, Sweden

WMO: 024640

Lat: 59.354N

Lon: 17.942E

Elev: 47

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.0	10.4	1.1	5.8	5.7	5.4	7.3	12.9	23.0	34.2	20.8	34.3	3.2	300	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	81.1	64.6	77.8	63.2	74.9	61.9	67.5	76.1	65.6	73.6	63.9	71.8	8.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	91.2	70.5	62.6	85.0	69.0	60.7	79.4	67.6	31.9	76.2	30.4	73.4	29.2	71.9	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.8	16.6	14.8	DB	-0.4	84.8	7.0	3.7	-5.4	87.4	-9.5	89.5	-13.4	91.6	-18.5	94.2	(4)
			WB	-0.9	69.4	6.9	2.5	-5.9	71.2	-10.0	72.7	-13.8	74.1	-18.9	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.0	28.3	28.3	33.5	42.3	51.4	59.0	64.8	62.9	54.7	44.9	37.3	31.7
	DBStd	14.48	8.59	8.58	7.27	6.21	6.43	5.18	5.04	4.98	5.45	6.30	6.73	8.41
	HDD50	3245	672	608	513	241	61	2	0	0	20	181	381	566
	HDD65	7398	1137	1028	977	680	424	189	67	99	311	625	830	1031
	CDD50	1428	0	0	0	11	103	273	458	401	160	22	0	0
	CDD65	107	0	0	0	0	1	10	60	35	1	0	0	0
	CDH74	827	0	0	0	0	20	101	483	220	3	0	0	0
	CDH80	139	0	0	0	0	0	10	100	29	0	0	0	0
Wind	WSAvg	7.2	7.5	7.3	7.4	7.4	7.3	7.3	6.7	6.4	6.8	7.0	7.3	7.5
Precipitation	PrecAvg	21.7	1.5	1.1	1.0	1.2	1.3	2.0	2.8	2.5	2.2	2.0	2.0	1.9
	PrecMax	31.5	2.9	2.8	2.7	3.4	3.1	4.4	7.6	6.4	6.7	5.4	4.1	4.6
	PrecMin	14.8	0.2	0.2	0.0	0.2	0.3	0.1	0.7	0.1	0.5	0.4	0.6	0.4
	PrecStd	3.7	0.7	0.6	0.6	0.7	0.7	1.0	1.7	1.4	1.4	1.2	0.9	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.8	46.6	56.7	67.7	76.8	81.1	86.9	83.2	73.3	61.2	53.4	50.9
		MCWB	42.8	42.4	46.1	52.5	58.8	62.0	66.0	65.9	61.6	55.8	50.4	47.2
	2%	DB	42.9	43.6	51.4	62.5	72.0	76.9	82.7	79.6	69.7	58.7	50.4	46.6
		MCWB	40.7	40.1	42.9	50.2	56.8	60.7	65.2	65.1	60.0	54.5	47.9	44.0
	5%	DB	40.8	41.2	47.1	58.6	68.1	73.6	79.2	75.9	66.3	56.4	48.4	44.3
		MCWB	38.8	38.2	40.8	48.3	55.1	58.9	64.0	63.9	58.5	52.9	46.4	42.3
	10%	DB	38.9	39.2	43.3	54.0	64.0	70.2	76.0	73.0	63.7	54.0	46.6	42.0
		MCWB	37.1	36.9	38.5	44.8	52.2	57.6	62.7	62.4	56.9	51.0	44.8	40.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.4	43.4	46.7	54.4	62.1	65.9	70.7	70.4	63.9	57.6	51.1	47.5
		MCDB	44.8	45.6	54.3	63.6	71.9	76.6	79.5	78.1	69.5	60.0	52.6	50.2
	2%	WB	40.8	40.6	43.6	51.0	58.7	63.0	68.0	67.4	61.1	55.2	48.4	44.4
		MCDB	42.1	42.8	49.6	60.7	67.9	72.8	78.2	74.5	67.0	57.7	49.9	45.9
	5%	WB	38.7	38.6	41.3	48.6	56.0	60.8	66.0	65.6	59.7	53.6	46.6	42.1
		MCDB	40.1	40.6	45.8	57.1	65.7	69.5	74.8	72.6	64.8	56.0	48.2	43.6
	10%	WB	37.0	36.9	39.3	46.0	53.7	59.0	64.3	63.9	58.0	51.7	44.9	40.2
		MCDB	38.4	38.8	42.8	52.8	62.4	67.5	73.0	70.9	62.5	54.0	46.3	41.5
Mean Daily Temperature Range	5% DB	MDBR	8.4	10.0	12.8	16.9	17.8	16.4	16.3	15.8	14.2	11.0	7.9	7.9
		MCDBR	7.8	9.7	17.8	24.0	24.1	22.7	22.3	20.3	17.1	12.1	9.4	8.1
		MCWBR	7.3	7.6	11.4	13.6	12.2	10.3	8.4	8.7	9.3	8.8	8.5	7.6
	5% WB	MCDBR	7.8	8.7	16.0	21.6	21.2	19.0	18.3	17.3	14.9	10.5	8.8	7.7
		MCWBR	7.6	7.3	10.9	12.7	11.7	9.5	7.9	8.2	8.8	8.5	8.2	7.5
Clear-Sky Solar Irradiance	taub	0.284	0.322	0.340	0.377	0.374	0.364	0.392	0.382	0.349	0.337	0.310	0.269	
	taud	2.216	2.281	2.351	2.319	2.393	2.452	2.396	2.416	2.494	2.414	2.254	2.140	
	Ebn at Noon	163	217	252	261	272	276	264	258	249	212	149	121	
	Edh at Noon	14	22	28	34	34	32	33	30	24	19	14	10	
All-Sky Solar Radiation	RadAvg	95	281	761	1267	1674	1862	1719	1359	881	387	124	63	
	RadStd	14	45	86	121	143	107	162	122	96	62	28	7	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-43
Regional (52 neighbors)		+0.67	N/A	N/A	N/A	N/A	N/A	-209	-261	N/A	N/A

Nomenclature: See separate page